PRACTITIONER KNOWLEDGE REPOSITORY

MODULE 6

KVA, ColVA & Capital Costs

Completing the XVA Waterfall

The Final Two Floors: The XVA waterfall was structurally complete after CVA, DVA, FVA, and MVA — but two costs were missing. KVA is the rent on regulatory capital. ColVA is the value of your right to post the cheapest collateral. Without them, you don't know the true cost of the building.

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Series Trail — Final Module

M0	Greeks & Sensitivities
M1	Basel Accord Evolution
M2	FRTB
M3	CVA / DVA
M4	FVA / MVA
M5	ISDA, Netting & Collateral
M6	**KVA, ColVA & Capital Costs ◀ THIS MODULE ◀ SERIES COMPLETE**
MD	Practitioner's Dictionary — forthcoming

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OVERVIEW

Module 6 — What We Cover

1 **Setting the Scene**
XVA waterfall recap — what was still missing and why

2 **ColVA: Plain English**
CTD intuition, who pays, vs FVA distinction

3 **ColVA: Full Derivation**
Formula, single-CCY example, multi-CCY worked example (EUR/USD/GBP)

4 **KVA: Plain English**
Rent on capital — intuition, pre/post Basel III contrast

5 **KVA: Conceptual Framework**
Building blocks, formula term-by-term, hurdle rate

6 **KVA: Full Derivation**
SA-CCR inputs, G-SIB surcharge, full worked example

7 **Complete XVA Waterfall**
All six components, interactions, three collateral regimes

8 **XVA Desk P&L Attribution**
P&L sources, hedgeability, IFRS 13, inception-to-maturity case

9 **Regulatory & Strategic**
Basel IV output floor, FRTB CVA, client selection, future of XVA

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The XVA Waterfall — What Was Still Missing

CVA

✓ Covered M3–M5

Counterparty default risk

DVA

✓ Covered M3–M5

Own default benefit

FVA

✓ Covered M3–M5

Funding cost of uncollateralised exposure

MVA

✓ Covered M3–M5

IM posting funding cost

ColVA

◀ THIS MODULE

CTD optionality — value of cheapest collateral choice

KVA

◀ THIS MODULE

Rent on regulatory capital — THIS MODULE

ColVA — The Cost of Having a Choice

ColVA in one sentence:

The value of the right to post whichever eligible collateral asset is cheapest on any given day under a multi-currency CSA.

01 Collateral Spread

The difference between eligible collateral rates (EUR OIS vs USD OIS vs GBP OIS). Wider spread = more valuable option.

02 Spread Volatility

Higher volatility of relative rates = more time value in the CTD option. Long-dated trades benefit most.

03 Correlation

Correlation between trade value and collateral spread — wrong-way risk can amplify or reduce ColVA.

BOND FUTURES ANALOGY

CTD in ColVA is identical to CTD in bond futures: the short party delivers the cheapest bond from the deliverable basket. The poster delivers the cheapest eligible collateral from the CSA basket. Both options have value.

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ColVA Formula — The Spread Option

Single Eligible Currency (Baseline)

```
ColVA = 0   (no optionality)
```

No choice = no option value. ColVA is zero by construction.

General Multi-Asset Case

```
ColVA = -∫ EPE(t) × {E[min_i(r_i(t))] - r_base(t)} × df(t) dt
```

Requires multi-dimensional spread modelling (n×n correlation matrix across all eligible currencies). Computed via Monte Carlo in practice.

Two-Currency Case

```
ColVA = -∫ EPE(t) × E[max(s_ij(t),0)] × df(t) dt
```

`EPE(t)`	Expected Positive Exposure — collateral amount outstanding
`s_ij(t)`	Collateral spread: rate_i(t) − rate_j(t)
`E[max(s,0)]`	Spread call option value (zero strike)
`df(t)`	Discount factor to present value
`− sign`	Cost to receiver; benefit to poster

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ColVA — Multi-Currency CSA: EUR + USD + GBP

Parameter	Value
Notional / Tenor	EUR 100M | 5Y IRS
EUR OIS (base)	3.20%
USD OIS + x-ccy basis	5.30% − 0.25% basis = 5.05% EUR-equivalent
GBP OIS + x-ccy basis	5.10% − 0.35% basis = 4.75% EUR-equivalent
Spread vol (EUR/USD)	40 bps p.a.
Spread vol (EUR/GBP)	35 bps p.a.

Year	EPE (EUR M)	E[max(s,0)]	ColVA Contrib
1	2.1	0.9 bps	−1,890
2	3.8	1.9 bps	−7,220
3	4.2	3.2 bps	−13,440
4	3.6	4.4 bps	−15,840
5	2.4	5.4 bps	−12,960
TOTAL ColVA			**−EUR 51,350**

ColVA = −EUR 51,350 ≈ −0.5 bps of notional per year | EUR cash remains CTD today but spread volatility creates future switching value | ColVA grows with tenor, vol, and current spread proximity to zero

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KVA — The Rent on Regulatory Capital

KVA in one sentence:

The present value of the cost of holding regulatory capital against a derivative trade over its entire life.

Before Basel III

- Low RWA: Basel II IMM produced thin capital charges for diversified books
- Cheap equity: ROE targets 20%+, capital perceived as abundant
- Competition: banks priced on spread — charging KVA meant losing trades

After Basel III

- CET1_req 13%+ for G-SIBs: min + CCB + G-SIB surcharge + SCB
- SA-CCR: higher, more stable RWA — less model optimisation
- Capital now scarce: 12% hurdle on EUR 100M trade = EUR 156K/yr cost

WAREHOUSE ANALOGY

Each trade occupies warehouse space (capital). The rent rate is the hurdle. Total rent = space × rate × time. KVA is the present value of all future rent payments.

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KVA Formula — Term by Term

$$KVA = h \times \int_0^T E[K(t)] \times df(t)\, dt$$

h

Hurdle Rate

Required annual return on equity. Typically 10–15% for G-SIBs. Set by the board. The 'rent rate' per unit of capital.

E[K(t)]

Expected Capital

Average regulatory capital consumed by the trade at each future date. E[K(t)] = E[RWA(t)] × CET1_req. Follows the EPE profile.

df(t)

Discount Factor

Converts future capital costs to present value. Future rent payments are worth less today — standard time value of money.

$\int_0^T \ldots dt$

Integration

Sum all future capital costs across the trade's entire life. Captures the full time profile — not just today's capital charge.

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G-SIB Surcharge — Structure & KVA Impact

Bucket	CET1 Surcharge	Representative Banks (2025)	KVA vs Non-G-SIB
1	1.0%	StanChart, UniCredit, Barclays, BNP, Deutsche	+14%
2	1.5%	BofA, Goldman, Morgan Stanley, UBS, Crédit Agricole	+21%
3	2.0%	Citi, HSBC	+29%
4	2.5%	JPMorgan Chase	+36%
5 (vacant)	3.5%	Deterrent buffer — no bank currently assigned	N/A

EUR 182K
Non-G-SIB baseline

EUR 208K
G-SIB Bucket 1
+14%

G-SIB Bucket 2
+21%

EUR 247K
G-SIB Bucket 4
+36%

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KVA — EUR 100M 5Y IRS, BBB, G-SIB Bucket 2

Step-by-Step Calculation

1 SA-CCR EAD

Eff.Notional = 100M × 4.42y × 0.50% = EUR 2.21M
EAD = 1.4 × (0 + 2.21M) = EUR 3.09M

2 RWA

RWA = EAD × RW(BBB 100%) = EUR 3.09M

3 Capital K_0

K_0 = 3.09M × 8.5% CET1_req = EUR 262,990

4 E[K(t)] Profile

Follows EPE — peaks year 2–3, declines to zero at maturity

5 KVA

h × Σ E[K(t)] × df(t) × Δt = 12% × EUR 411,300

KVA = EUR 49,356

≈ 0.5 bps/yr × EUR 100M notional

Parameter Change	KVA Impact
Hurdle: 10% → 15%	±EUR 20,565 per 1%
BBB → AA (RW 20%)	−80% → EUR 9,871
Bucket 2 → non-G-SIB	−18% → EUR 40,615
5Y → 10Y tenor	~×2 — more capital years
Bilateral → CCP	~zero — no CCR RWA

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Full XVA Decomposition — Three Regimes

XVA	No CSA	Full VM (EUR Cash)	Multi-CCY CSA (5% TH)
CVA	**−285,000**	**~zero**	−58,000
DVA	+42,000	~zero	+8,500
FVA	**−195,000**	**~zero**	−39,000
MVA	zero	**−28,400**	**−28,400**
ColVA	zero	zero	**+51,350**
KVA	−49,356	−8,200	−22,000
TOTAL XVA	**−487,356**	**−36,600**	**−87,550**

No CSA: −EUR 487K
CVA+FVA dominate. 10× more expensive than fully collateralised.

Full VM: −EUR 37K
MVA dominates. 92% total cost reduction.

Multi-CCY: −EUR 88K
ColVA offsets 37% of remaining cost.

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XVA Interactions — Double-Count Risks

CVA vs FVA

CVA ↔ FVA

A defaultable counterparty provides a funding benefit — if they default, you don't need to fund the exposure. Computing both independently double-counts this benefit. Solution: FCA/FBA split — FVA computed net of CVA.

FVA vs ColVA

FVA ↔ ColVA

The CTD asset chosen under ColVA affects the funding cost of the remaining uncollateralised exposure. Both must be modelled jointly in consistent simulation paths — not summed independently.

MVA vs KVA

MVA ↔ KVA

IM reduces SA-CCR EAD (PFE multiplier) → reduces RWA → reduces KVA. But IM must be funded → increases MVA. Net effect depends on relative sizes. Must be modelled together.

All six XVA components must be computed in a jointly consistent simulation framework — not independently summed. Ignoring interactions overstates total XVA and misprices trades.

XVA P&L — Sources, Hedgeability & IFRS 13

XVA	Primary P&L Driver	Hedgeable?	IFRS 13 Status
CVA	Credit spread moves (daily); default events (discrete)	Partially — CDS hedges delta	Mandatory under IFRS 13
DVA	Own CDS spread; own credit deterioration	Controversial — own-credit paradox	Mandatory under IFRS 13
FVA	Funding spread moves; CSA addition events	Partially — internal funding book	Included by most dealers; IASB no guidance
MVA	SIMM recalibration; market vol moves; netting changes	Partially — swaption vol hedge	Included; IFRS 13 Level 3
ColVA	Collateral spread moves; CTD switching events	Yes — x-ccy basis swaps	Included; Level 2/3
KVA	RWA changes; regulatory updates; hurdle revision	Very limited	Most banks EXCLUDE — not fair value

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Lifecycle P&L — EUR 100M IRS, Inception to Maturity

Event	CVA	FVA	MVA	KVA	Net XVA
Day 1: booked, no CSA	−285,000	−195,000	0	−49,356	**−529,35**
Yr 1: cpty spreads tighten 20bps	+52,000	0	0	+4,200	**+56,20**
Yr 2: CSA added (full VM)	+260,000	+190,000	−28,400	+41,156	**+462,75**
Yr 3: SIMM recalib +10%	0	0	−2,840	0	**−2,84**
Yr 5: trade matures	+8,600	+5,000	+31,240	+8,200	**+53,04**
NET OVER LIFE	+35,600	+0	+0	+4,200	**+39,80**

CSA addition in Year 2 is the dominant P&L event — releasing CVA and FVA reserves against the new collateralised valuation. CSA renegotiation events are the largest single XVA P&L trigger in a derivatives book.

Basel IV, FRTB CVA & Strategic Implications

Basel IV Output Floor

- RWA floor = 72.5% of SA RWA — constrains IMM benefit for derivative books
- SA-CCR already high vs IMM → floor binds less for SA users
- KVA increases as floor raises effective RWA across the book
- Accelerates shift to clearing — CCP trades get more favourable RWA treatment

Strategic Implications of KVA Pricing

Product Pricing

Long-dated uncollateralised + high-RW = structurally expensive. Capital-adjusted pricing hierarchy emerges.

Client Selection

AA/AAA (RW 20%) structurally preferred to BBB (RW 100%). KVA shapes relationship banking strategy.

Balance Sheet Opt.

Clearing, compression, netting, tenor reduction — all reduce RWA and therefore KVA. Direct economic incentive.

FUTURE OF XVA

AI/ML for real-time XVA (sub-second at execution) | Digital asset ColVA (tokenised collateral CTD) | Climate capital add-ons (new KVA dimension for carbon-intensive counterparties)

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Module 6 — Key Takeaways

01

ColVA = value of the CTD option in a multi-currency CSA. Single-currency CSA → ColVA = 0. Multi-currency → spread option value, grows with volatility and tenor.

02

ColVA formula: $-\int EPE(t) \times E[\max(s_{ij},0)] \times df(t)\, dt$. The spread option is valued at zero strike — always positive expected value when vol > 0.

03

$KVA = h \times \int E[K(t)] \times df(t)\, dt$. The rent on regulatory capital. Driven by hurdle rate, RWA profile, G-SIB surcharge, and trade tenor.

04

G-SIB surcharge creates a structural KVA disadvantage of 14–36% vs non-G-SIBs on the same trade. Explains systematic loss of capital-heavy business to regional banks.

05

XVA interactions are real: CVA/FVA double-count; FVA/ColVA interact through collateral choice; MVA/KVA have offsetting effects. Must be modelled jointly.

06

CSA addition is the largest single XVA P&L event in a derivatives book — releasing CVA and FVA reserves simultaneously. XVA desks monitor collateral agreements closely.

Module 6 — Acronyms & Short Forms

XVA & Valuation Adjustments	
XVA	X-Valuation Adjustment — collective term for all valuation adjustments
CVA	Credit Valuation Adjustment — cost of counterparty default risk
DVA	Debit Valuation Adjustment — own default benefit
FVA	Funding Valuation Adjustment — cost of funding uncollateralised exposure
MVA	Margin Valuation Adjustment — cost of posting Initial Margin
ColVA	Collateral Valuation Adjustment — value of CTD optionality in multi-CCY CSA
KVA	Capital Valuation Adjustment — cost of regulatory capital over trade life
CTD	Cheapest-to-Deliver — lowest-cost eligible collateral asset
EPE	Expected Positive Exposure — average future exposure across scenarios
ENE	Expected Negative Exposure — average future negative exposure
PFE	Potential Future Exposure — SA-CCR add-on component
RC	Replacement Cost — SA-CCR current exposure component

Capital & Regulatory	
CET1	Common Equity Tier 1 — highest quality regulatory capital
RWA	Risk-Weighted Assets — capital base for minimum ratio calculation
EAD	Exposure at Default — SA-CCR output; RWA = EAD × Risk Weight
SA-CCR	Standardised Approach for Counterparty Credit Risk
G-SIB	Global Systemically Important Bank — subject to capital surcharge 1–3.5%
SCB	Stress Capital Buffer — US dynamic buffer derived from DFAST stress test
CCB	Capital Conservation Buffer — fixed 2.5% CET1 add-on under Basel III
FRTB	Fundamental Review of the Trading Book — Basel trading book capital reform
IMM	Internal Model Method — CCR model approach, subject to Basel IV output floor
AAD	Adjoint Algorithmic Differentiation — computes all Greeks in one backward pass

Collateral & Market	
CSA	Credit Support Annex — ISDA collateral agreement governing VM and IM
VM	Variation Margin — daily MtM settlement collateral
IM	Initial Margin — MPoR buffer; segregated under UMR
MPoR	Margin Period of Risk — 10 days bilateral, 5 days cleared
UMR	Uncleared Margin Rules — mandatory bilateral IM phased 2016–2022
SIMM	Standard Initial Margin Model — ISDA sensitivity-based IM calculation
OIS	Overnight Index Swap — risk-free rate used to discount collateralised trades
IRS	Interest Rate Swap — vanilla fixed-for-floating derivative
CCP	Central Counterparty — clearing house; reduces bilateral CCR
x-ccy	Cross-currency — refers to cross-currency basis or swap
ROE	Return on Equity — hurdle rate benchmark for KVA calculation

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Series Complete — What Comes Next

✓ **SERIES COMPLETE**

Basel I–IV → FRTB → CVA/DVA → FVA/MVA → ISDA, Netting & Collateral → KVA, ColVA & Capital Costs

Coming Next — Module D: Practitioner's Dictionary

📖 Three-Layer Entries

Every term: plain English definition + formula (where applicable) + practitioner usage note

🎨 Illustrated

Key structural concepts rendered as visuals — netting sets, XVA waterfall, ISDA hierarchy, collateral flows

🔗 Cross-Referenced

Terms link to the module where they were introduced — acts as an index to the full series

📊 ~80+ Terms

All terms seeded across M0–M6 plus cross-cutting regulatory and market structure entries

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